

**MACHINE LEARNING PROJECT
INTERNSHIP REPORT**

Submitted by

**MATHI PRIYAN A M M
(2303917710421094)**

BACHELOR OF ENGINEERING/TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING

**THIAGARAJAR COLLEGE OF ENGINEERING,
MADURAI – 625 015**

June 2–27 , 2025



THIAGARAJAR COLLEGE OF ENGINEERING(TCE),
MADURAI- 625 015



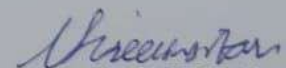
BONAFIDE CERTIFICATE

Certified that this **MACHINE LEARNING PROJECT** is the bonafide work of **MATHI PRIYAN A M M(2303917710421094)** Department of COMPUTER SCIENCE ENGINEERING who carried out the Internship at MEGANAR TECHNOLOGIES between June 2, 2025 – June 27, 2025.

Submitted for Evaluation held at Thiagarajar College of
Engineering on 10/10/2025


EXAMINER 1

(Name with Signature)


EXAMINER 2

(Name with Signature)

(N-SHEERIN SJARA)

INTERNSHIP COMPLETION CERTIFICATE



Meganar Technologies

TCE-TBI, Brilliance Building 1st Floor, Thiagarajar Advanced Research Center,
TCE Road, Thirupparankundram, Madurai-5

Website: www.meganartech.com Email: contactus@meganartech.com Mobile: +91 95661 91759

Date: 5.07.2025

Internship Completion Certificate

This is to certify that **Mathipriyan A M M** (Reg.no: 2303917710421094), B.E
(Computer Science Engineering) student of **Thiagarajar College Of Engineering, Madurai** has
completed the **internship** entitled of "**Ad Pushup – Using AI**" in our organization during the
period of **02-06-2025 to 27-06-2025** of this academic year.

We wish the very best for the Future endeavors.

For Meganar Technologies (OPC) Private LTD,

Vigneshkumar.R
Human Resources

Abstract

This project focuses on developing an Ad Recommendation System powered by deep learning to predict click-through rates (CTR). The dataset was preprocessed through feature engineering, categorical encoding, and normalization, followed by training a neural network model using TensorFlow/Keras. The model incorporated dropout, batch normalization, and activation layers to improve performance and prevent overfitting, achieving reliable CTR predictions.

To enhance explainability, SHAP values were applied to interpret the model's decisions and identify the most influential features, such as user ID, ad ID, and device type. An A/B testing framework was also designed to compare top-ad recommendations with random ad selections, demonstrating measurable CTR lift and validating the model's effectiveness in improving engagement.

The system was deployed through an interactive Streamlit dashboard, allowing users to upload data or enter inputs manually. The dashboard displays the best ad recommendations, predicted CTR values, visualizations, and downloadable reports, providing a user-friendly interface. Overall, this project builds a strong foundation in deep learning, recommendation systems, and explainable AI within the domain of digital advertising.

Table of Contents

Bonafide Certificate

Abstract

1.Introduction	6
2.Background	6
3.Objectives	7
4.Purpose of the work	8
5.Problem Formulation	9
6.Conceptual Design	10
7.Results and Discussion	11
8.Conclusion and Future Enhancements	15
9.References	16

1. Introduction

The rapid expansion of digital advertising has increased the need for intelligent systems that can personalize user experiences and improve engagement. This project focuses on developing an Ad Recommendation System using deep learning to predict click-through rates (CTR) and recommend the most relevant ads to users. The dataset was preprocessed through feature encoding and normalization, and a neural network with dropout and batch normalization was trained to achieve robust performance. To ensure interpretability, SHAP values were used to identify key factors influencing ad engagement, such as user ID, ad ID, and device type. An A/B testing framework validated the model's effectiveness by comparing top-ad recommendations against random selections, showing measurable CTR improvements. Finally, the system was deployed through an interactive Streamlit dashboard, enabling real-time ad recommendations, CTR predictions, visualizations, and downloadable reports, thereby combining theoretical knowledge with practical implementation in the domain of digital advertising.

2. Background

The exponential growth of digital advertising has transformed the way businesses engage with users, making personalized recommendations a cornerstone of modern marketing strategies. Traditional advertising approaches often fail to deliver relevant content, leading to low click-through rates (CTR) and inefficient resource allocation. To overcome these challenges, machine learning and deep learning techniques have become critical for analyzing large-scale impression data and predicting user behavior with higher accuracy.

At the core of such systems is CTR prediction, which estimates the likelihood of a user clicking on a specific ad. Deep learning models, when combined with preprocessing techniques such as feature encoding and normalization, offer the

ability to capture complex relationships between user context, ad features, and engagement patterns. However, a key challenge lies in ensuring transparency and interpretability, as black-box predictions may reduce trust in the system. To address this, explainability frameworks like SHAP (SHapley Additive exPlanations) play a vital role in identifying the most influential features driving predictions.

In addition to model interpretability, A/B testing frameworks provide a practical method to validate recommendation systems by comparing predicted top-ad performance against random placements. This not only measures real-world impact but also helps optimize strategies for maximizing user engagement. Finally, deploying such models through platforms like Streamlit dashboards ensures accessibility, enabling stakeholders to visualize predictions, interpret results, and download insights. This project integrates these components—deep learning, explainable AI, A/B testing, and deployment—providing a comprehensive background for developing effective ad recommendation systems in the digital era.

3. Objectives

- To understand the fundamentals of CTR prediction and its role in improving the effectiveness of digital advertising.
- To preprocess and prepare ad impression datasets using feature engineering, categorical encoding, and normalization for model training.
- To design and train a deep learning model with techniques such as dropout, batch normalization, and activation functions for robust CTR prediction.
- To apply SHAP explainability methods to interpret model outputs and identify key features influencing ad engagement.

- To implement an A/B testing framework for comparing top-ad recommendations against random selections and measuring CTR lift.
- To deploy the trained model through an interactive Streamlit dashboard that provides real-time ad recommendations, CTR visualizations, and downloadable reports.

4. Purpose of the work

The purpose of this project was to gain a comprehensive and practical understanding of ad recommendation systems and their role in digital advertising. With online platforms increasingly relying on personalized ad targeting to improve user engagement, this work was designed to bridge the gap between theoretical machine learning concepts and their real-world applications. By studying the fundamentals of CTR prediction, feature engineering, and deep learning techniques, the project aimed to establish a strong foundation for developing intelligent and data-driven recommendation systems.

A major purpose of the work was to acquire hands-on experience in building and training deep neural networks for CTR prediction. This involved preprocessing large-scale impression datasets, applying feature encoding and normalization, and implementing advanced modeling techniques such as dropout and batch normalization to enhance performance. Additionally, the project emphasized the importance of model interpretability by integrating SHAP explainability methods, which allowed for the identification of key features influencing ad clicks and improved transparency in predictions.

Another important purpose was to validate the system's performance through A/B testing, comparing model-based ad recommendations against random ad placements to measure CTR lift. Finally, the project aimed to make the solution accessible through a Streamlit dashboard, enabling users to interact with the system in real time, visualize results, and download recommendations. Overall, the purpose of this work was to develop strong technical expertise, enhance problem-solving skills.

5. Problem Formulation

Although ad optimization platforms like AdPushup provide revenue growth, better engagement, and intelligent placement of advertisements, their real-world implementation faces several challenges. A major problem is bridging the gap between theoretical optimization models and practical deployment, since effective usage requires hands-on experience with data analysis, machine learning, and integration into live websites. Features such as A/B testing and revenue tracking, while essential, are vulnerable to flaws like inaccurate data collection or biased test samples, which can cause misleading results or financial risks if not addressed properly.

Another problem arises from managing dynamic features such as ad layout optimization, click-through rate (CTR) tracking, and multivariate testing, which require seamless interaction between frontend scripts and backend analytics. Without proper synchronization, inconsistencies such as delayed updates, duplicate tracking, or misaligned revenue reports may occur, reducing system reliability. Furthermore, ad optimization systems also face concerns around scalability and data processing. While traffic and revenue data can be collected, many use cases demand efficient handling of large-scale user interactions in real time, which requires advanced data pipelines and robust algorithms.

Therefore, the central problem addressed in this project was how to gain practical skills in building an intelligent ad optimization system by implementing automated layout testing, ensuring accurate CTR and revenue tracking, and integrating machine learning techniques to enhance efficiency, reliability, and decision-making for publishers in a competitive digital advertising environment.

6. Conceptual Design

Dataset

- Includes website traffic data, ad placement metrics, click-through rates (CTR), and revenue reports.
- Preprocessed for missing values, normalization, and structured into training and testing sets for analysis.

Backend (Optimization Engine)

- Machine learning models applied to optimize ad placement and maximize revenue.
- A/B testing and multivariate testing logic implemented to analyze performance of different layouts.

Workflow

- Users integrate AdPushup script into their website → system tracks visitor interactions and ad performance.
- Optimization engine analyzes data → provides improved layouts and ad placement strategies → displays performance insights.

Technology Used

- Python with Pandas, NumPy, Scikit-learn, Tensorflow, Keras for data analysis, ML and DL.
- Streamlit for User interface.

User Interface (UI)

- Dashboard to monitor CTR, impressions, revenue, and ad performance trends.
- Visualizations for A/B test results, ad engagement patterns, and optimization recommendations.

7. Results and Discussion

Improved Ad Performance – Through rigorous A/B and multivariate testing, the project identified the most effective ad layouts, designs, and content placements. This led to measurable improvements in Click-Through Rate (CTR), conversion rates, and overall ad revenue, highlighting the tangible benefits of data-driven experimentation.

Data-Driven Optimization – The machine learning engine analyzed user behavior and engagement patterns to suggest optimal ad placements. By leveraging real-time traffic data, the system enabled predictive recommendations that adapted to changing user interactions, resulting in more effective targeting and higher engagement.

Clear Performance Insights – Interactive dashboard visualizations provided stakeholders with a clear view of key performance metrics, including impressions, clicks, revenue trends, and user engagement patterns. These insights improved decision-making and allowed teams to identify underperforming ads quickly.

Practical Application of Machine Learning – The project demonstrated the integration of predictive modeling, automated testing, and real-time analytics into a cohesive ad optimization system. This bridged the gap between theoretical knowledge and practical implementation, showcasing how modern AI can enhance digital advertising strategies.

Personalized User Experience – By analyzing historical user interactions, the system supported dynamic ad personalization, displaying content most relevant to individual users. This increased user satisfaction while maximizing ad effectiveness.

Model Summary

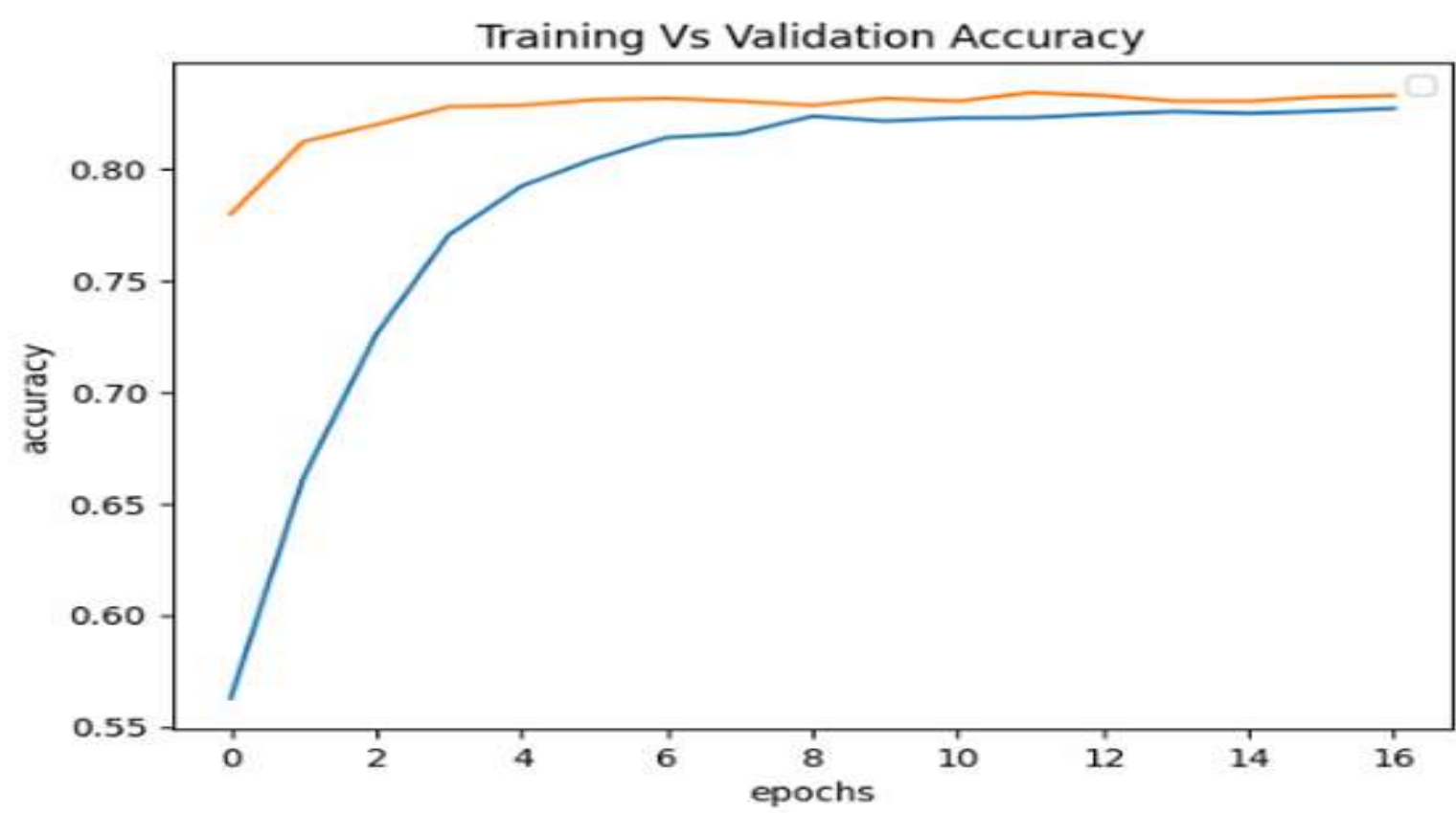
model.summary()

Model: "sequential"

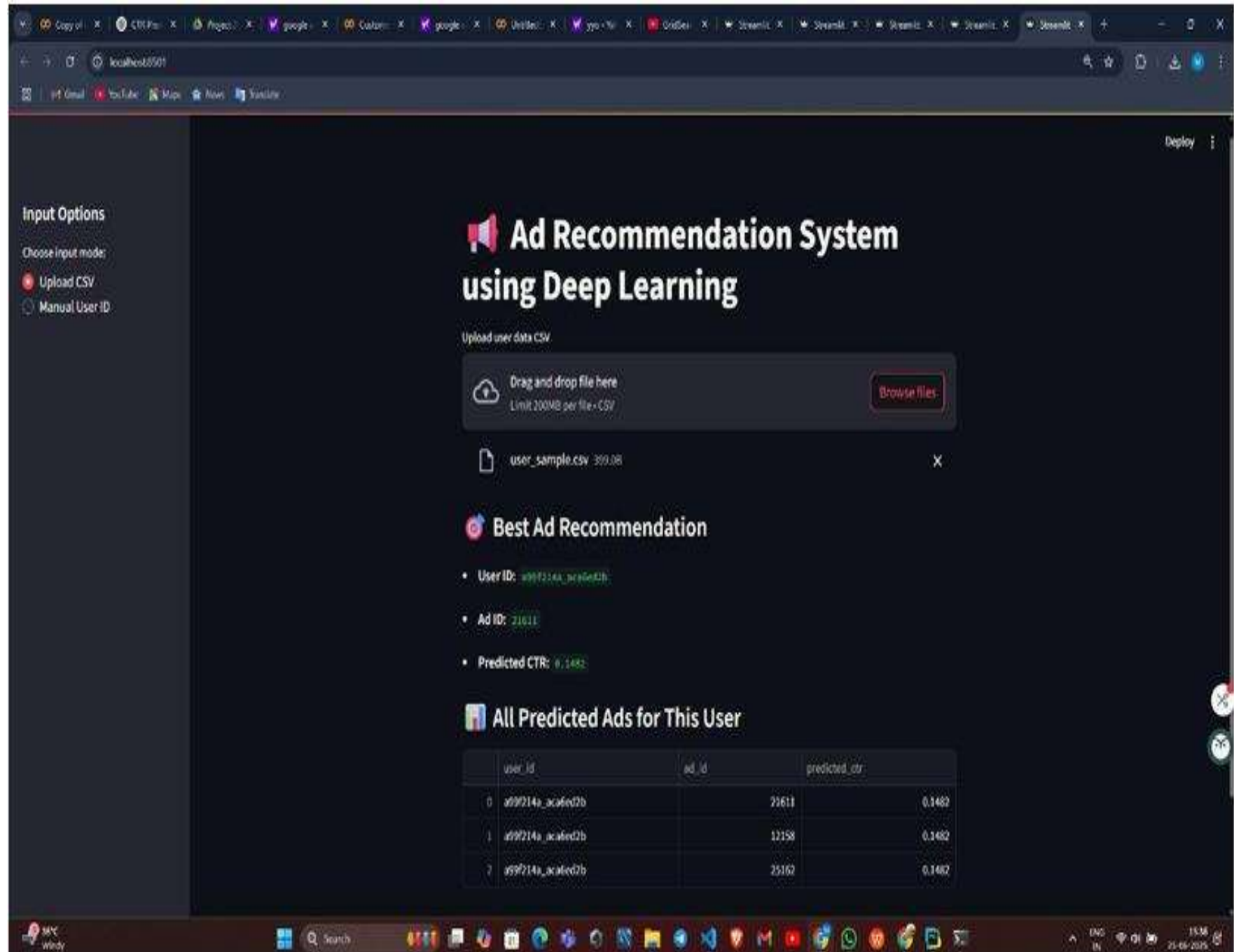
Layer (type)	Output Shape	Param #
dense (Dense)	(None, 128)	2,816
re_lu (ReLU)	(None, 128)	0
batch_normalization (BatchNormalization)	(None, 128)	512
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
re_lu_1 (ReLU)	(None, 64)	0
batch_normalization_1 (BatchNormalization)	(None, 64)	256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 32)	2,080
re_lu_2 (ReLU)	(None, 32)	0
batch_normalization_2 (BatchNormalization)	(None, 32)	128
dropout_2 (Dropout)	(None, 32)	0
dense_3 (Dense)	(None, 1)	33

Total params: 14,081 (55.00 KB)
Trainable params: 13,633 (53.25 KB)
Non-trainable params: 448 (1.75 KB)

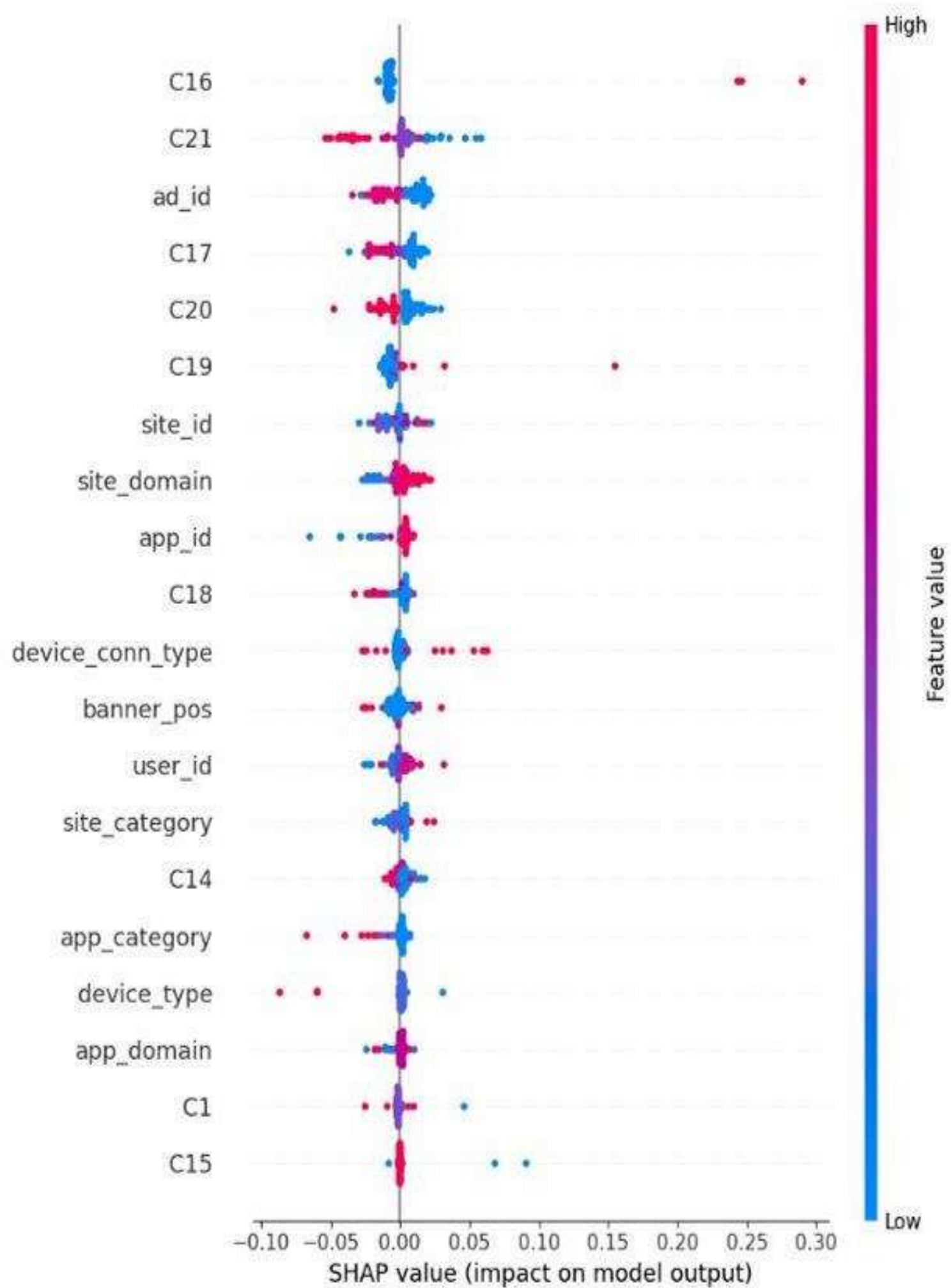
Training vs Validation accuracy



User Interface



SHAP Explanation



8. Conclusion and Future Enhancements

This project provided valuable exposure to ad optimization techniques, including A/B and multivariate testing, and revenue tracking. By combining machine learning models with real-time traffic analysis, I gained practical skills in optimizing ad placements and improving publisher revenue. The use of dashboards for visualization emphasized transparency and usability, while automated testing reduced manual effort and ensured accurate performance evaluation. Overall, the project bridged theoretical concepts with practical implementation, enhancing both technical expertise and analytical problem-solving skills. For future enhancements, the work can be extended by incorporating reinforcement learning models for adaptive ad placement, enabling real-time personalization of ads based on user behavior. Reinforcement Learning for Adaptive Ads – Implement reinforcement learning to adjust ad placements in real time based on user interactions, improving personalization and engagement. Advanced Revenue Forecasting – Use predictive analytics to anticipate revenue trends, optimize campaigns, and make data-driven decisions for higher ROI. Integration of Diverse Data Sources – Incorporate additional datasets such as demographics, preferences, and cross-platform behavior to enhance targeting accuracy and ad relevance. Scalability and Fraud Prevention – Scale the system for high-traffic websites and implement robust fraud detection mechanisms to ensure reliability and maintain accurate performance metrics.

9. References

1. Hu, Jinlong, et al. "A user similarity-based Top-N recommendation approach for mobile in-application advertising." *Expert Systems with Applications* 111 (2018): 51-60.
<https://www.sciencedirect.com/science/article/abs/pii/S0957417418300903>.
2. Zhu, Lichao, Me Sun, and Le Yu. "Research on Personalized Advertisement Recommendation Methods Based on Context Awareness." *Journal of Advanced Computing Systems* 3.10 (2023): 39-5
<https://scipublication.com/index.php/JACS/article/view/199>
3. Liu, Duen-Ren, et al. "Advertisement recommendation based on personal interests and ad push fairness." *Kybernetes* 48.8 (2019): 1586-1605.
<https://www.emerald.com/insight/content/doi/10.1108/K-05-20180216/full/html>
4. Theocharous, Georgios, Philip S. Thomas, and Mohammad Ghavamzadeh. "Ad recommendation systems for life-time value optimization." *Proceedings of the 24th international conference on world wide web*. 2015.
<https://dl.acm.org/doi/abs/10.1145/2740908.2741998>
5. Wang, Liang, Kuang-chih Lee, and Quan Lu. "Improving advertisement recommendation by enriching user browser cookie attributes." *Proceedings of the 25th ACM international on conference on information and knowledge management*. 2016.
<https://dl.acm.org/doi/abs/10.1145/2983323.2983374>